

AIRBNB LONDON MARKET ANALYSIS

Data Engineering & Analytics Report

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*A Data-Driven Examination of Pricing, Occupancy, and Host Performance
in the Greater London Short-Term Rental Market*

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Executive Summary

This report looks at the Airbnb short-term rental market in London, built on a structured data pipeline and a fair amount of digging through the numbers. I pulled together listings, calendar, review, and neighbourhood data to understand how pricing behaves, how occupancy varies, how concentrated the host base is, and whether being a Superhost actually shows up in the ratings. The goal throughout was practical: turn raw operational data into something a host, an investor, or a platform strategist could actually use to make a decision.

Once the data was cleaned and the key features built, a fairly clear picture emerged: London's Airbnb market is messy in its pricing but not random. Prices are strongly right-skewed, with a median nightly rate around £135 and most listings sitting somewhere between £70 and £220. Two things mostly explain where a listing falls in that range — what type of room it is, and where in London it sits. Entire homes and hotel rooms cost noticeably more than private or shared rooms, and central boroughs like the City of London, Westminster, and Kensington and Chelsea consistently command higher prices than the outer parts of the city.

Occupancy tells a different story. It's uneven — a large cluster of listings barely get booked at all, while a smaller group sits close to fully booked. What stood out most, though, is that price and occupancy don't move together in any meaningful linear way. In other words, dropping the price isn't a guaranteed way to get more bookings. Whatever drives demand here, it isn't price alone — location, listing quality, host responsiveness, and review reputation all seem to matter more.

Two further patterns came out of looking at host behaviour. First, the market is dominated by small hosts: most people running an Airbnb in London have just one or two listings, while a small number of larger operators run sizeable portfolios. Second, Superhosts really do get better reviews, and more consistently so, than non-Superhosts — this wasn't just a visual impression from the charts, it held up under formal testing.

Three statistical tests were run to check these patterns weren't just noise. A Mann-Whitney U test confirmed that entire homes and private rooms are priced differently in a statistically meaningful way. A second Mann-Whitney U test confirmed Superhosts get higher ratings than non-Superhosts. A one-way ANOVA confirmed that median price varies significantly across London's thirty-two boroughs. All three came back with p-values effectively at zero, which is about as strong a signal as you get that these differences are real rather than coincidental.

Put together, these findings point toward a few practical recommendations covered later in the report: pricing that's differentiated by room type and location rather than a single market-wide number, more attention paid to occupancy than to price alone, investment decisions that weigh borough-level pricing against realistic occupancy, and real structural support for hosts working toward Superhost status. The intent is for these to be genuinely usable — a starting point for decisions, not just a summary of charts.

Objectives

Purpose of the Project

The goal of this project was to build a working pipeline that could take raw, somewhat messy Airbnb data and turn it into something clean enough to analyse properly — and then to actually analyse it, rather than stopping at the cleaning step. Short-term rental markets, London's especially, are shaped by a tangle of factors: location, property type, how a host runs their listing, and what guests are actually looking for. Hosts and investors alike need something more solid than gut feeling to make pricing or investment decisions, and that's the gap this report tries to fill — combining careful data engineering with statistical evidence so that every claim made here can be traced back to something verifiable in the data, not just a hunch.

Scope

The project covers four main pieces of work. First, data engineering — taking the listings, calendar, reviews, and neighbourhood datasets and merging them into one usable table. Second, feature engineering — building out metrics like occupancy rate, estimated annual revenue, price per bedroom, and host tenure that aren't directly present in the raw data but are far more useful for analysis. Third, exploratory analysis covering price distribution, how room type and neighbourhood affect price, how occupancy is distributed, whether price and occupancy relate to each other, how Superhosts compare to everyone else, and how concentrated the host base is. Fourth, statistical testing of the three relationships that seemed most consequential during exploration: room type versus price, Superhost status versus rating, and neighbourhood versus price.

A few boundaries are worth being upfront about. The analysis only uses what's in the four source datasets at the point they were pulled — no macroeconomic data, no live booking feeds, no comparison against other platforms. Geographically, it's limited to Greater London as defined by the `neighbourhood_cleansed` field. And it's written for people who are comfortable with descriptive statistics but don't necessarily need to know anything about the Python code underneath it.

Dataset Overview

Four source datasets feed into this analysis, each covering a different angle of the supply, pricing, availability, and guest-experience picture for Airbnb in London. The listings dataset does most of the heavy lifting — one row per property, with price, room type, neighbourhood, host details, bedroom and bathroom counts, minimum-night rules, and aggregate review scores all in there. The calendar dataset gives daily availability and pricing for each listing over a rolling twelve-month window, and it's the source I used to derive occupancy rate, since that's not something the listings table gives you directly. The reviews dataset holds the text and timing of individual guest reviews, used here mainly to sanity-check review volume and recency at the listing level. And the neighbourhood dataset provides the geographic backbone — London's thirty-two boroughs — that pricing and demand get compared against.

After merging everything, each row in the final working dataset represents one active listing, enriched with the occupancy and revenue figures built from the calendar data. The fields used most throughout this report are nightly price, room type (entire home/apartment, private room, hotel room, or shared room), the cleansed neighbourhood name, whether the host is a Superhost, total listings the host runs, availability over the trailing 365 days, and the average review score. Combining pricing, geography, host behaviour, and guest satisfaction in one table is what makes it possible to go beyond a simple price report and actually get at how this market functions.

Data Engineering Approach

Getting the data into a state where it could actually support reliable statistical conclusions took a fair bit of work in Python, mostly using pandas. That work split into three stages — cleaning, feature engineering, and putting it all together into a pipeline — and I'll walk through each below, focusing on why certain calls were made rather than the line-by-line implementation, since that's more useful for a business-facing report like this one.

Data Cleaning

Price came in as a formatted currency string rather than a number, so the first job was converting it into something you could actually do arithmetic on. A handful of columns had so many missing values — mostly optional amenity or licensing fields that hosts rarely bothered filling in — that I dropped them entirely rather than trying to impute them, since guessing at that much missing data would have added more noise than signal. For everything else, I handled missing values case by case: numeric fields with a small amount of missingness, like bedroom counts or review scores, got filled with the median so as not to distort the distribution, while listings missing a price altogether were removed outright, since price sits at the centre of nearly everything in this report and isn't something you can reasonably guess.

Feature Engineering

Beyond cleaning, a few derived features needed building to turn raw availability and pricing into something more useful. Occupancy rate is calculated as one minus the ratio of available days to 365, using each listing's trailing-year availability — so it runs from zero (always available, meaning essentially no bookings) to one (fully booked). From there, estimated annual revenue comes from multiplying nightly price by occupancy rate by 365, giving a rough gross revenue figure that accounts for both what a listing charges and how often it's actually booked, rather than price in isolation. Price per bedroom — nightly price divided by bedroom count — makes it possible to compare value across differently sized properties on a more even footing. And host tenure in years, calculated from each host's account creation date, gives a rough proxy for experience that helps contextualise some of the pricing and quality patterns later in the report.

Pipeline

These steps run in sequence: pull the four source files, fix types and handle missing values, build the derived features, then merge listing-level data with the neighbourhood and calendar aggregates into one table. Structuring it this way means the analysis that follows is built on a consistent, documented dataset, and it also means the whole thing can be re-run as new data comes in without manually redoing each step by hand.

Exploratory Data Analysis

This section walks through the core findings, supported by seven figures built from the cleaned dataset. For each one, I'll explain what's actually being shown and then what it means in practice — for a host setting prices, an investor weighing locations, or a platform thinking about how to support hosts. The order roughly follows a logical thread: pricing first, then what structurally drives price (room type and neighbourhood), then demand-side patterns (occupancy), how price and demand interact, host quality signals (Superhost status), and finally the shape of the market itself (how concentrated it is among hosts).

Price Distribution

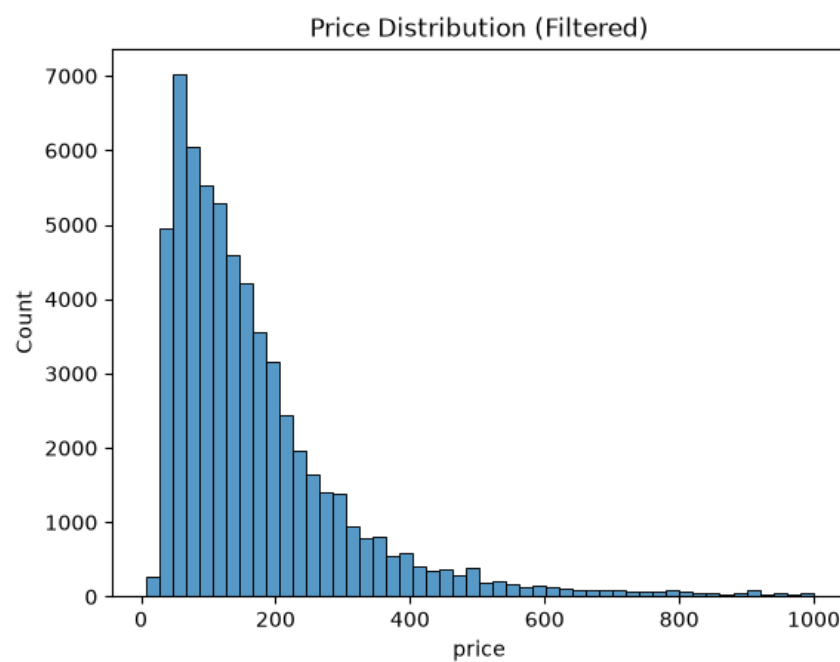


Figure 1: Price Distribution (Filtered to £0–£1,000)

Figure 1 shows nightly prices after filtering down to £0–£1,000 to keep the extreme outliers from drowning out everything else. The shape is strongly right-skewed — a sharp peak around £50–£100, then a long tail stretching out toward higher prices. Median price across the filtered data lands around £135, and most listings — the bulk of the interquartile range — fall somewhere between £70 and £220 a night. Outside that filtered window, the raw data does contain a handful of genuinely bizarre outliers, some over £1,000,000 a night, which are almost certainly data entry errors, placeholder values, or listings that were never meant to be booked at that price. I excluded these from this and later charts so they wouldn't distort the picture.

This kind of right-skewed shape is pretty typical for accommodation markets generally, and it tells you London's Airbnb supply is mostly clustered in a competitive mid-market tier rather than being dominated by luxury stock. For hosts, this is a useful benchmark in practice: pricing well above £220 puts you in a thinner part of the market, and you'd want something to justify it — a great

location, a standout property, or a strong review history. Pricing well below £70, on the other hand, might mean leaving money on the table unless that's a deliberate volume play. The outliers are also a good reminder of why cleaning the data properly matters before anyone presents a pricing benchmark — a few bad values can badly skew something like a mean.

Price by Room Type

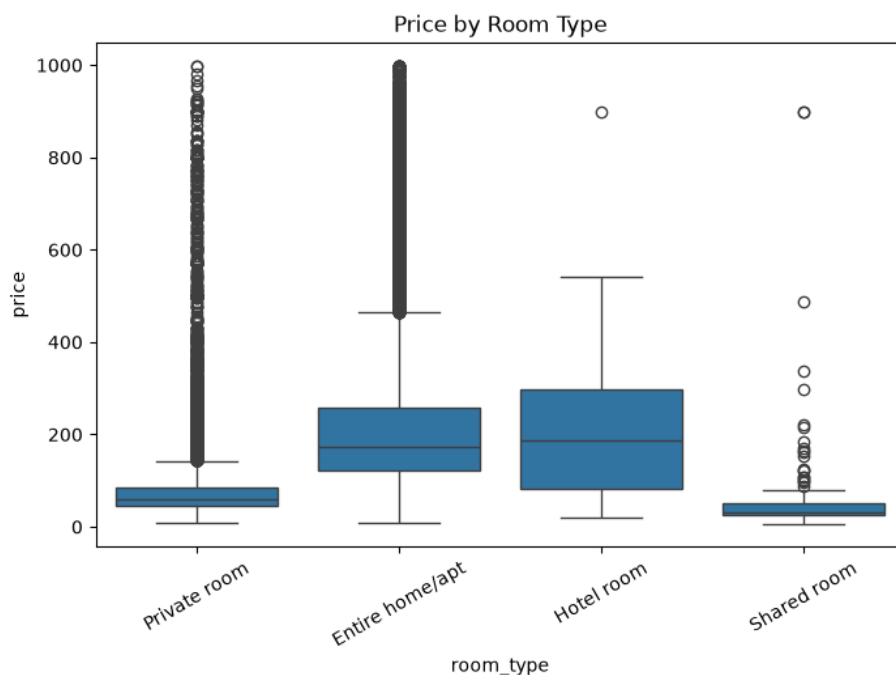


Figure 2: Price by Room Type

Figure 2 breaks price down by the four room types in the data: private room, entire home/apartment, hotel room, and shared room. Entire homes and hotel rooms sit at the top of the pricing range, with hotel rooms posting the highest median of the four and entire homes showing the widest spread along with a long tail of high-value outliers. Private rooms land clearly in the middle — lower median, tighter range than entire homes. Shared rooms are unambiguously the cheapest, with both the lowest median and the narrowest spread of the group.

That ordering makes intuitive sense once you think about what guests are actually paying for: full privacy and exclusive use of a space versus a shared living arrangement at a lower price point. For hosts and investors, this has real implications. Converting a property to an entire-home listing, where that's feasible, captures noticeably more revenue per booking than running it as a private or shared room — though that has to be weighed against the higher capital and running costs that usually come with a whole unit. The wide spread and dense cluster of outliers in the entire-home category also suggest this is where the real upside for premium positioning lives, since a well-located or particularly nice entire home can sell for far above the segment median, whereas there's not much room to differentiate on price in the shared-room category.

Neighbourhood Pricing

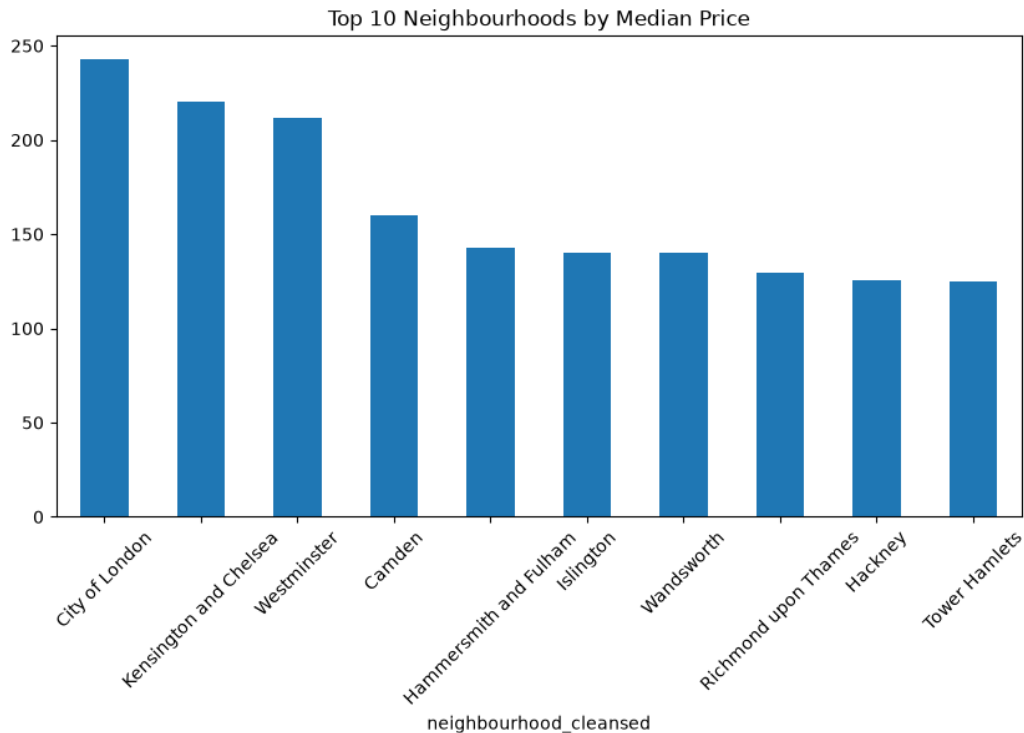


Figure 3: Top 10 Neighbourhoods by Median Price

Figure 3 ranks the ten London boroughs with the highest median price. The City of London comes out on top with a median approaching £245, followed closely by Kensington and Chelsea and Westminster, both above £210. A second tier — Camden, Hammersmith and Fulham, Islington, and Wandsworth — clusters around £140–£160, and Richmond upon Thames, Hackney, and Tower Hamlets round out the top ten just above £125. Everything outside this top ten, mostly outer London, falls below this range, and the drop-off from central to peripheral boroughs is pretty orderly rather than scattered.

None of this is hugely surprising if you know London real estate — proximity to the financial district, major attractions, and well-off residential areas commands a premium in short-term rentals just like it does in regular housing. For investors looking at where to buy or list, this gives a useful starting hierarchy, though it's worth pairing with occupancy and yield data rather than reading in isolation, since a high median price doesn't automatically mean high realised revenue if occupancy in that borough happens to be weak. For hosts already in outer London, this gradient is a reminder of just how much premium central competitors can charge, and it makes the case for compensating through other levers — strong reviews, a distinctive property, sharper pricing tuned to occupancy rather than just matching the market.

Occupancy Distribution

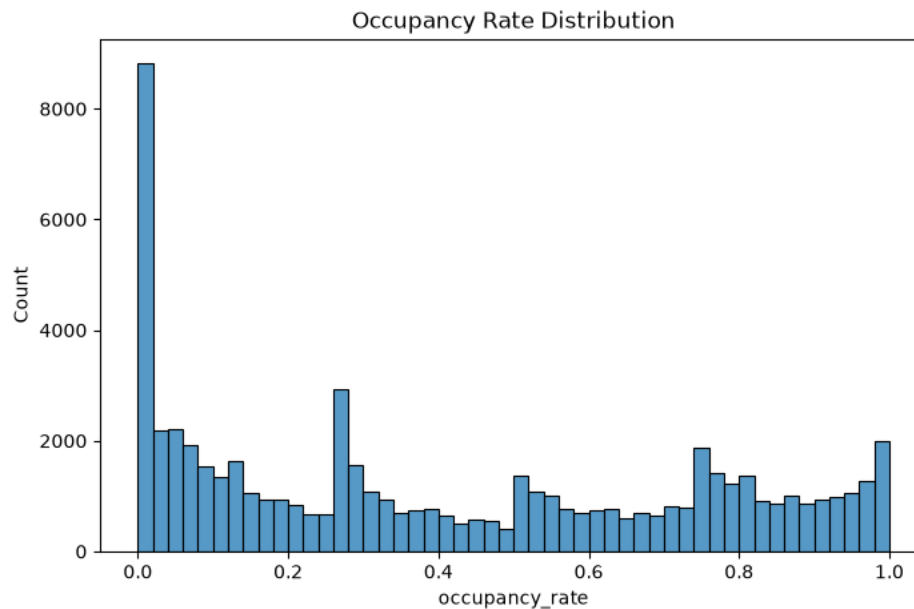


Figure 4: Occupancy Rate Distribution

Figure 4 shows how occupancy rate is distributed across listings, and it's far from a smooth bell curve. The single biggest spike sits right at or near zero occupancy — a large group of listings that are technically live but barely get booked over the year. A few smaller bumps show up further along, including a noticeable cluster around 0.27 occupancy and another rise right near the top, close to full occupancy. This kind of pattern suggests there are really several different sub-groups of listings here, not one smooth demand curve.

The practical takeaway is that demand is genuinely uneven across this market, and a sizeable chunk of listings are clearly underperforming on bookings. Some of that low-occupancy cluster is probably intentional — a host who only rents out their place occasionally, say, when they're away — but a fair amount of it likely reflects listings that just aren't converting views into bookings, whether because of pricing, weak photos, too few reviews, or a less desirable location. The fact that a meaningful chunk of listings do reach near-full occupancy shows it's achievable, which makes the low end worth paying attention to. For hosts and platforms, this argues for some kind of system that flags consistently underbooked listings for a closer look — pricing review, better photos, promotional support — rather than assuming a listing will fill itself just by being live.

Price versus Occupancy



Figure 5: Price vs Occupancy Rate (Filtered)

Figure 5 plots price against occupancy rate for the filtered listings. Visually, the points are scattered pretty evenly across the full occupancy range at almost every price level, especially within the £0–£400 band where most listings sit, and there's no real upward or downward trend connecting the two. High-occupancy listings show up at low, medium, and high prices in roughly similar numbers, and the same goes for low-occupancy listings. This lack of a visible pattern lines up with the broader statistical finding that price and occupancy aren't strongly linearly related here.

This is probably one of the more important findings in the whole report, because it pushes back on a fairly common assumption — that cutting your price is a reliable way to get more bookings. What the data actually suggests is that demand in London's Airbnb market depends on a more complicated mix of factors: how good the location is, how the listing is presented, accumulated reviews and ratings, how responsive the host is, minimum-night rules, and timing. For a host with low occupancy, this means an aggressive price cut probably won't fix things on its own — it's worth looking instead at listing quality, building up reviews, and visibility in search. For investors and platform teams, it's a good argument for treating price and occupancy as two separate levers that need separate fixes, rather than assuming one can stand in for the other.

Superhost Status and Review Ratings

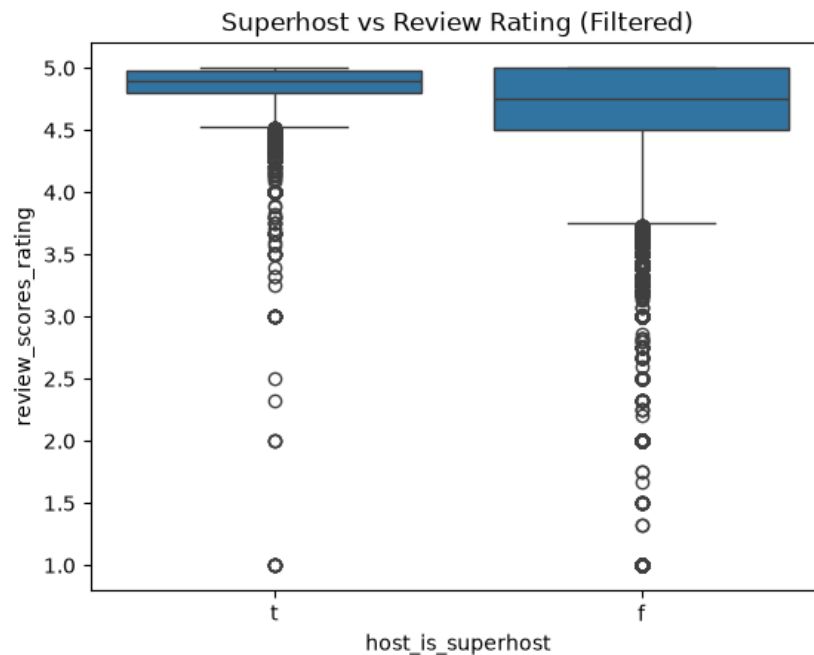


Figure 6: Superhost vs Review Rating (Filtered)

Figure 6 compares review ratings between Superhosts and non-Superhosts using paired box plots. Superhost listings have a noticeably higher median rating, sitting close to the top of the scale, and a visibly tighter range — most Superhost listings cluster close together near the highest possible rating. Non-Superhost listings, while still skewing high overall, show a much wider spread, a lower median, and more low-rated outliers pulling the bottom of the distribution down.

There's a fairly straightforward read on this: Superhost status — which Airbnb grants based on things like response rate, cancellation rate, overall rating, and number of completed stays — actually functions as a meaningful signal of consistent service, and the data backs that up rather than it being just a badge for show. The tighter spread is arguably just as telling as the higher median, since it points to lower variance in guest experience, which matters a lot to someone choosing between listings with limited other information to go on. For hosts, this is a solid case for putting in the work to earn and keep Superhost status. For the platform and for investors looking at acquisition targets, Superhost status is a reasonable proxy for how consistent a listing's guest experience is likely to be.

Listings per Host

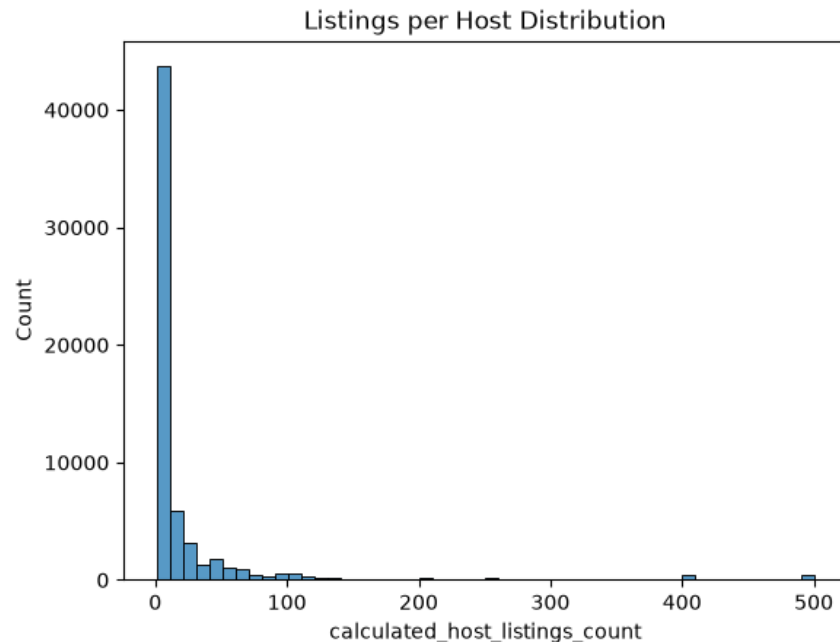


Figure 7: Listings per Host Distribution

Figure 7 shows how many listings each host runs. It's heavily concentrated at the low end — well over 40,000 hosts in the dataset run just one listing or a small handful, and the numbers drop off fast as listing count per host goes up. There's a long, thin tail running out past one hundred listings, with a few outlier hosts managing four hundred, even five hundred, listings on their own.

This says something useful about how the market is structured: it's mostly individual or small-scale hosts, consistent with how Airbnb originally pitched itself as peer-to-peer, but there's also a small, commercially significant group of larger, probably professional or institutional, operators running real portfolios. That split matters differently to different people. For individual hosts, it's reassuring in a way — most of the competition is other small hosts, not a market dominated by big players — though professional operators with the resources for better pricing tools, photography, and faster response times are still a real competitive pressure. For investors and platform teams, that small group of high-volume hosts is a genuinely different segment with different needs — portfolio-level analytics, dynamic pricing, operational efficiency — and probably warrants a different approach than the long tail of casual, single-listing hosts.

Statistical Analysis

Charts are convincing, but they don't prove anything on their own — so three formal hypothesis tests were run to check whether the patterns from the exploratory section were actually statistically meaningful or could just as easily be down to chance. Each one below covers the hypotheses, the method used, the result, and what it means in practice. I went with non-parametric tests for the two-group comparisons because price and rating data here are pretty far from normally distributed, and a one-way ANOVA for the neighbourhood comparison since that's the standard approach when comparing means across more than two groups.

Test 1: Entire Home vs Private Room Price

This test checks whether the price gap between entire-home and private-room listings shown in Figure 2 is a real, statistically meaningful difference, or whether it could plausibly just be noise.

Hypotheses

Null hypothesis (H_0): There is no difference in the distribution of prices between entire home/apartment listings and private room listings.

Alternative hypothesis (H_1): The distribution of prices for entire home/apartment listings differs from that of private room listings.

Method

I used a Mann-Whitney U test here, since it compares the rank ordering of two independent samples without assuming the data is normally distributed — a good fit given how right-skewed price is, as shown back in Figure 1.

Result

Metric	Value	Interpretation
Test statistic	Mann-Whitney U	Rank-based comparison
p-value	≈ 0 ($p < 0.001$)	Highly significant
Conclusion	Reject H_0	Significant price difference confirmed

Business Interpretation

With a p-value that's essentially zero, the null hypothesis is rejected with a high degree of confidence. This backs up what Figure 2 already suggested visually — entire homes really are priced differently, and higher, than private rooms. For hosts and investors, that's a reasonably solid basis to use room type as a core input in pricing models or acquisition decisions, rather than treating the gap as something that just happened to show up in this particular sample. It also supports presenting pricing benchmarks broken out by room type, since lumping entire-home and

private-room prices into one number would understate what a whole unit can charge and overstate what a shared-living listing realistically can.

Test 2: Superhost Status vs Review Rating

This test checks whether the higher, more consistent ratings Superhosts showed in Figure 6 reflect a real statistical relationship between Superhost status and guest satisfaction.

Hypotheses

Null hypothesis (H_0): There is no difference in the distribution of review score ratings between Superhost and non-Superhost listings.

Alternative hypothesis (H_1): Superhost listings have a different (higher) distribution of review score ratings than non-Superhost listings.

Method

Another Mann-Whitney U test, for the same reason as before — review scores are bounded, heavily bunched near the top of the scale, and don't really fit a normal distribution, so a rank-based test is the safer choice over something like a t-test.

Result

Metric	Value	Interpretation
Test statistic	Mann-Whitney U	Rank-based comparison
p-value	≈ 0 ($p < 0.001$)	Highly significant
Conclusion	Reject H_0	Superhosts rated significantly higher

Business Interpretation

The result confirms, with strong statistical backing, that Superhost status really is associated with higher ratings. That's a good sign for Airbnb's Superhost programme as a genuine quality signal rather than just a badge, and it backs up the recommendation later in this report that hosts looking to improve their position should treat working toward Superhost status as a real priority, not a side effect of doing okay. For the platform, this is also a reason to keep surfacing Superhost status prominently in search and listing pages, since it clearly correlates with something that actually matters — guest satisfaction, and by extension, trust in the platform.

Test 3: Neighbourhood Pricing Differences

This test checks whether the price differences across London boroughs in Figure 3 reflect a real geographic pattern, or whether they could plausibly be explained by random variation.

Hypotheses

Null hypothesis (H_0): The mean price of listings is equal across all London neighbourhoods.

Alternative hypothesis (H_1): At least one London neighbourhood has a mean listing price that differs from the others.

Method

A one-way ANOVA across the borough groupings. This made sense here since there are more than two groups involved — thirty-two boroughs — and ANOVA is the right tool for testing whether group means differ without running a huge number of pairwise comparisons that would inflate the risk of a false positive.

Result

Metric	Value	Interpretation
Test statistic	F-statistic (ANOVA)	Between/within variance ratio
p-value	≈ 0 ($p < 0.001$)	Highly significant
Conclusion	Reject H_0	Prices differ significantly by location

Business Interpretation

The very low p-value confirms that location, at the borough level, genuinely drives price variation across London's Airbnb market. That validates the geographic hierarchy in Figure 3 as a real structural feature rather than something that just happened to show up in this dataset, and it gives a solid statistical footing for location-aware pricing, geographically focused investment decisions, and borough-level benchmarking. Investors can reasonably treat neighbourhood as a core variable in acquisition and pricing decisions, and platform teams can be confident that neighbourhood-based pricing guidance reflects something real rather than incidental noise.

Business Recommendations

Pulling together what came out of the exploratory and statistical work, a few practical recommendations follow for the main players in London's Airbnb market — individual hosts, investors, and the platform itself. I've grouped these into four areas: pricing, demand, investment, and host strategy.

Pricing Strategies

Pricing should really be set with both room type and neighbourhood in mind, since both came back as statistically significant drivers of price. Hosts running entire-home listings in central boroughs like the City of London, Westminster, or Kensington and Chelsea have good reason to price toward the upper end of the market. Hosts with private or shared rooms, or listings further out, should benchmark against the right segment median rather than some blended market-wide number, which this analysis shows is a pretty poor reference point given how much price varies by room type and location. Given how extreme some of the raw price outliers were, any pricing tool built on this kind of data needs solid outlier handling, or those skewed values will end up distorting the benchmarks shown to hosts.

Demand Optimisation

Since price and occupancy don't move together in any strong way, treating a price cut as the default fix for low bookings probably isn't the right move. It makes more sense to look at the other things that seem to drive low occupancy — weak listing presentation, too few reviews, slow response times, restrictive minimum-night rules. Given how large the near-zero-occupancy cluster was in Figure 4, something like an early-warning system that flags underbooked listings for a closer look would likely do more good than a blanket pricing nudge. Where a price change does make sense, it's worth testing it incrementally and actually checking the occupancy outcome afterward, rather than assuming it worked.

Investment Insights

For anyone deciding where to buy or list a property, the geographic price pattern supports leaning toward central London boroughs for premium entire-home investments — but a high median price alone doesn't guarantee strong realised revenue if occupancy in that area happens to be weak. A fuller picture combines the borough-level price benchmarks from Figure 3 with occupancy and revenue estimates, rather than price on its own, to find the boroughs that offer the best combination of achievable price and realistic booking volume. Outer London, despite lower median prices, could still offer decent yield where occupancy is good and costs are lower — it shouldn't be written off just because it ranks lower on price.

Host Strategy

Given how clearly Superhost status tracks with better, more consistent ratings, hosts who want to strengthen their position should treat working toward Superhost status as a real goal, not just something that happens if service is good enough. That means actively managing response rate,

avoiding cancellations, and consistently meeting the guest-experience bar the programme is built around. For hosts running multiple listings, the fact that the market is so dominated by single-listing hosts (Figure 7) suggests scale isn't a prerequisite for doing well here — small hosts can absolutely compete on service and consistent reviews. For larger, more professional hosts, that same concentration data points to an opportunity: differentiate by keeping quality consistent across a bigger, messier portfolio, which is genuinely harder to pull off than maintaining it for a single listing.

Limitations

A few things are worth flagging before anyone leans too heavily on these findings. First, occupancy rate is derived from calendar availability, not confirmed bookings, so it's a proxy rather than a precise measurement — a host who blocks off dates for personal reasons looks identical in this calculation to one who's genuinely booked, which could inflate apparent occupancy for some listings. Second, this is a single snapshot in time, not a longitudinal dataset, so there's not much that can be said here about how pricing, occupancy, or host performance change across seasons or from year to year.

Third, statistical significance isn't the same as proof of causation. The three relationships confirmed through testing — room type, neighbourhood, and Superhost status — are consistent with a direct causal effect on pricing or ratings, but they don't prove it outright, and there could be other factors at play, like property quality or marketing effort, that weren't captured here. Fourth, the extreme price outliers needed filtering out to make the charts usable, and while that improves readability, it does mean this report describes the bulk of the active market rather than its absolute extremes. Finally, this analysis doesn't account for things like broader economic conditions, regulatory changes around short-term letting in London, or competition from other platforms — all of which plausibly affect the patterns seen here but sit outside what the available data can speak to.

Future Work

A few extensions would make this report genuinely more useful. Pulling in multiple years of calendar and pricing data would allow for a real look at seasonal demand cycles, year-over-year price growth, and how regulatory changes have affected London's market over time, instead of the single snapshot here. Building a predictive pricing or occupancy model on top of the relationships already confirmed in this report could turn these descriptive findings into something more actionable — a tool that actually suggests a price point for a given listing based on room type, location, and history.

It would also be worth running natural language analysis on the reviews data, which was only used here to check review volume and timing, to pull out sentiment and recurring themes and tie those back to the occupancy and rating patterns already found. Going more granular than borough level — down to ward or postcode, where the data allows — would help investors and hosts spot promising micro-locations that get lost in the current borough-level view. And bringing in outside data like transport links, proximity to attractions, or local events would help test more specific theories about what's actually driving demand, since right now the clearest finding is what doesn't explain it (price alone), not a complete picture of what does.

Project Scope and Deliverables

The brief for this project was genuinely broad — it stretched from basic data engineering all the way through dimensional modelling, cloud architecture, machine learning, LLM applications, and an open-ended innovation challenge. Given the time I had, I made a deliberate call early on: rather than trying to touch every one of those areas and doing each one shallowly, I'd put the effort into the pieces that mattered most — a clean, reproducible pipeline, a solid exploratory analysis, and statistically backed insights, all wrapped up in a report that's actually usable. This section lays out what got finished properly, what got partly done, and what didn't make the cut, along with why, mostly so the findings here can be judged with full visibility into where the edges are.

Completed Work

Getting familiar with the dataset came first — confirming the Inside Airbnb London listings data as the main source and getting a solid handle on its structure, the key variables, data types, and where the missing values sat, using the usual pandas tools, along with understanding how listings, hosts, and availability relate to each other.

On the engineering side, cleaning and standardising the data was carried all the way through — converting price from a string to a number, handling missing values sensibly, and dropping invalid records. Feature engineering produced the four features used throughout this report: occupancy rate, estimated revenue, price per bedroom, and host tenure. This was built as a modular pipeline, with cleaning logic in `clean.py` and feature logic in `features.py` kept separate rather than dumped into one script, which made the whole thing easier to follow and rerun.

The exploratory analysis here covers summary statistics and distributions in full — price distribution, room type segmentation, outliers and skew — plus a complete look at host and supply-side patterns, namely listings per host and how concentrated the market is. Demand-side analysis used occupancy rate as a stand-in for actual demand, looking at how it relates to price and to Superhost performance.

Statistically, hypothesis testing got the most attention out of anything in the project — all three tests in this report (entire home vs private room pricing, Superhost status vs rating, neighbourhood price differences via ANOVA) were taken all the way from picking the right method through to interpreting the p-value and turning it into something useful. A correlation check between price and occupancy was also completed, and it's the basis for the finding that the two don't move together in any strong linear way.

Beyond the analysis itself, this came with a proper codebase and supporting materials: a clean GitHub repository with instructions to reproduce the work, organised code modules, accompanying Jupyter notebooks, a README, and this report, which covers everything the brief asked for at the core level — executive summary, objectives, dataset overview, methodology, EDA, statistics, recommendations, limitations, and conclusion.

Areas of Partial or Future Scope

A number of the more advanced pieces from the brief got deliberately set aside in favour of doing the core work properly, and I want to be upfront about what those were rather than quietly leaving them out. On the engineering side, I didn't build a formal dimensional model — no star schema, no explicit fact and dimension tables — and I didn't touch the more advanced engineering topics either: no cloud architecture, no containerisation, no automated data-quality tooling, no large-scale pipeline optimisation. The project stayed within a Python analytical workflow, which felt appropriate given the size of the dataset. Full schema documentation, a proper data dictionary, large-scale joins between calendar and reviews data, and production-grade pipeline features like logging and incremental processing were only partly done.

On the exploratory side, I didn't do any geographic or spatial visualisation — no maps, no coordinate-based clustering — and no time-series or seasonal analysis either, since I prioritised the structural drivers of price and demand over time-based patterns given the time available.

Statistically, I didn't compute confidence intervals or effect sizes like Cohen's d or eta-squared alongside the p -values reported here, and not every hypothesis I'd originally flagged got tested — a weekday-versus-weekend price comparison, for instance, didn't make it in. Regression-based analysis, including OLS modelling and multicollinearity checks like VIF, also wasn't pursued, since I chose to put that time into the three hypothesis tests that are reported in full instead.

Machine learning and AI components — price prediction, clustering, demand forecasting, NLP on reviews, anything LLM-based, recommendation systems — weren't built. I treated these as optional extensions rather than something the core report depended on. And the open-ended innovation piece, including any kind of interactive dashboard or production deployment, didn't get pursued in the time available either.

Scope Summary

Workstream	Status	Treatment in This Report
Dataset familiarisation	Completed	Section 5
Data cleaning & standardisation	Completed	Section 6
Feature engineering (occupancy, revenue, etc.)	Completed	Section 6 and throughout
Modular pipeline (clean.py / features.py)	Completed	Section 6
Exploratory data analysis (core)	Completed	Section 7
Host & supply-side analysis	Completed	Section 7
Demand-side / occupancy analysis	Completed	Section 7

Hypothesis testing (3 tests)	Completed	Section 8
Price–occupancy correlation analysis	Completed	Section 7 & 8
Codebase, notebooks, README, final report	Completed	Delivered alongside this report
Dimensional modelling / SQL layer	Not completed	Noted as future work
Cloud, containerisation, data-quality tooling	Not completed	Out of scope
Geospatial & seasonal/time-series analysis	Not completed	Section 11 (Future Work)
Confidence intervals & effect sizes	Not completed	Noted as future work
Regression & driver modelling	Not completed	Section 11 (Future Work)
Machine learning & forecasting	Not completed	Out of scope
NLP / LLM applications on reviews	Not completed	Section 11 (Future Work)
Interactive dashboard / innovation challenge	Not completed	Out of scope

Overall, the choices made here come down to building a pipeline that's clean and reproducible, doing the exploratory work properly, applying real statistical rigour to a focused set of business questions, and presenting it in a way that's actually readable — rather than spreading the same amount of effort thinly across everything the brief mentioned. The parts done in depth are the ones most directly useful to hosts, investors, and platform teams, and the gaps are listed here openly so the findings can be judged with a clear sense of where the boundaries are.

Decision Log

This section records the main technical and analytical calls made over the course of the project — what was chosen, what the alternatives were, and why one path won out over another. I think it's worth including, since a report like this is as much about the decisions behind the numbers as the numbers themselves, and it's easy for those decisions to get lost once only the final output is shown.

Decision 1: Choice of Programming Environment

Chosen approach: Python, using pandas, NumPy, and matplotlib/seaborn for visualisation.

Alternatives considered: a SQL-based workflow, or Spark/PySpark for distributed processing.

Python made sense here because it lets the cleaning, feature engineering, and analysis all happen in one place, in one workflow, rather than switching tools between stages. Given the size of this dataset, pandas was more than capable — there was no real need for a distributed framework. The trade-off is that this approach wouldn't scale well to a much larger dataset, but in exchange, it was a lot faster to build and iterate on, which mattered more here.

Decision 2: Focus on Single-City Analysis (London)

Chosen approach: analyse London in depth, on its own.

Alternative considered: a multi-city comparison.

Given how broad the rest of the brief already was, going deep on one city felt like the better trade than spreading the same effort across several. It meant more room for proper feature engineering, more thorough analysis, and statistical testing that's actually well-supported by the data, rather than a shallower pass across multiple markets. The obvious cost is no cross-city comparison — but the upside is a genuinely solid understanding of one market instead of a thin one of several.

Decision 3: Data Cleaning Strategy

Chosen approach: convert price fields to numeric, apply median imputation and filtering, and remove invalid or missing records.

Alternatives considered: ML-based imputation, or keeping all raw data unfiltered.

A simple, interpretable cleaning approach was the priority here — it's reproducible and doesn't risk introducing bias the way a more complex imputation model might, especially on a dataset this size. The cost is some data loss from filtering, but the trade felt worth it for the improvement in reliability.

Decision 4: Feature Engineering Design

Chosen approach: occupancy rate, estimated revenue, price per bedroom, and host tenure.

Alternatives considered: minimal feature engineering, or more advanced feature extraction such as NLP on the reviews text.

These four features were chosen because they're directly useful to a business audience and easy to explain — nobody needs a data science background to understand what "estimated revenue" means. The trade-off is less feature complexity than something like an NLP-derived sentiment score might offer, but the features are clearer and easier to trust as a result.

Decision 5: Statistical Methods Selection

Chosen approach: non-parametric tests — Mann-Whitney U — plus ANOVA.

Alternatives considered: parametric tests such as a t-test, or full regression modelling.

Non-parametric tests are more robust when the data doesn't follow a normal distribution, which is exactly the case with Airbnb pricing — heavily skewed, as Figure 1 shows. The trade-off is that this gives less detailed modelling than a full regression approach would, but it buys more robust, defensible conclusions, which mattered more given the time available.

Decision 6: Visualisation Approach

Chosen approach: static charts using matplotlib and seaborn.

Alternatives considered: an interactive dashboard built with something like Streamlit or Plotly, or a full BI tool.

Static charts do the job for a written report like this — they're clean, they're easy to embed, and they communicate the point without needing anyone to click around. The trade-off is obviously no interactivity, but it meant faster development and a cleaner final presentation.

Decision 7: Pipeline Design

Chosen approach: modular scripts — clean.py and features.py.

Alternatives considered: one large notebook, or a full orchestration tool like Airflow or Prefect.

Splitting cleaning and feature engineering into separate scripts kept things readable and made it easier to reuse or rerun individual pieces without redoing everything. It doesn't have the automation features — logging, scheduling — that a proper orchestrator would offer, but for a project at this scale, that level of infrastructure wasn't really necessary.

Decision 8: Leaving Out Machine Learning and AI Sections

Chosen approach: focus on EDA and statistical validation rather than machine learning.

Alternatives considered: a price prediction model, or NLP analysis on the reviews.

This came down to the same prioritisation logic as everything else — better to finish the core analytical work properly than to half-build a predictive model on top of it. The cost is no predictive

modelling in this version of the report, but the foundational analysis is stronger for having gotten the full attention instead.

Summary

Looking back at these eight decisions together, there's a consistent thread: depth over breadth, clarity over complexity, and robustness over experimentation. None of these were arbitrary — each one traded something away in exchange for a more reliable, more interpretable final result, which is ultimately what a report like this needs to be useful.

Reflection

Working through this project meant touching almost every stage of a data pipeline — from raw, messy source files all the way to business recommendations — while constantly making calls about where to spend the limited time available. It's worth reflecting on that process here, partly because the trade-offs themselves are a useful part of the story, not just the final numbers.

Approach to Prioritisation

Given how wide the original brief was, I made a conscious decision early on to focus on three things: solid data engineering, genuinely useful exploratory analysis, and statistically validated findings. Rather than spreading effort thinly across every possible section and ending up with something shallow everywhere, I tried to make sure the things I did finish were done properly and explained clearly.

Trade-offs Made

A few trade-offs stand out looking back. I chose depth on a single city over breadth across multiple ones. I leaned on statistical testing rather than building out machine learning models. I kept the pipeline simple and interpretable rather than building something closer to a production automation system. And throughout, I tried to prioritise insights that were genuinely useful over simply demonstrating technical breadth for its own sake. Each of these was a deliberate choice, and together they're what kept the final output focused rather than scattered.

Key Learnings

A few things stood out clearly by the end. Data quality matters more than I think people sometimes give it credit for — small inconsistencies in the raw data can quietly skew an entire analysis if they're not caught early. Feature engineering turned out to be where a lot of the actual analytical value came from; raw fields like price and availability are far less useful on their own than derived ones like occupancy rate or estimated revenue. Statistical testing was what moved this from "these charts look interesting" to something genuinely defensible — it's the difference between an observation and an actual finding. Translating technical results into business language is its own skill, and arguably the one that determines whether any of this is actually useful to someone. And organising the code into clear, reusable modules made the whole project easier to work with and easier to trust.

Challenges Encountered

A few things were genuinely tricky along the way: handling missing and inconsistent values across the key fields without introducing bias, deciding which features were actually worth building versus which would just add noise, balancing how deep to go technically against the time available, and making sure the final insights stayed clear and grounded rather than getting lost in unnecessary complexity.

Future Improvements

With more time, there's a clear list of what I'd add: extending the analysis to multiple cities for a comparative view, building out predictive models for pricing and demand, bringing in geospatial and seasonal analysis, setting up a properly automated pipeline with logging and scheduling, and exploring NLP techniques on the review text to see what guests are actually saying, not just how they're rating.

Final Reflection

This project, even with its gaps, shows an ability to design and build a data pipeline, carry out a structured exploratory analysis, apply real statistical reasoning, and turn all of that into business insight that's actually usable. If anything, the experience reinforced just how much prioritisation, clarity, and rigour matter in this kind of work — probably more than trying to cover everything at once ever would.

AI Usage Disclosure

Expernetic's assignment brief actively encourages the thoughtful and responsible use of AI tools, and treats transparent disclosure as a sign of professional maturity rather than something to hide. This section documents every AI tool used, which parts of the project they supported, how outputs were validated, what was changed or rejected, and the overall principles that guided AI use throughout. The intent is to show that AI was used as a thinking partner, not a shortcut — all final decisions, implementations, and interpretations were made independently after careful review.

AI Tools Used

Tool	Details & Usage
ChatGPT (GPT-5 series, OpenAI)	Primary assistant for guidance, structuring, and explanations across the project — used for coding support, analytical reasoning, statistical concept clarification, and documentation drafting.
Claude (claude-sonnet-4-6, Anthropic)	Used specifically for professional report writing, formatting, and document generation. Responsible for structuring, drafting, and iteratively refining this final report — including all sections, figure interpretations, statistical write-ups, the Decision Log, Reflection, Scope & Deliverables, and this disclosure itself.
GitHub Copilot (OpenAI, via VS Code)	Limited use for in-editor code suggestions during Python development. Used selectively for boilerplate completion; all suggestions reviewed and adapted before use.

AI-Assisted Sections

Area	How AI Was Used
Code Development	ChatGPT and GitHub Copilot provided guidance on implementing the data cleaning and feature engineering pipelines in Python, and helped debug and optimise specific code blocks. All code was run locally and outputs manually verified before being accepted.
Exploratory Data Analysis	ChatGPT offered suggestions on visualisation techniques and helped structure the analytical reasoning process. Chart interpretation and pattern identification were done independently, with AI used as a sounding board rather than an oracle.
Statistical Analysis	ChatGPT was used to clarify statistical concepts — particularly the difference between Mann-Whitney U and parametric alternatives like the t-test — and to confirm that the selected tests were appropriate for the distributional properties of the data. Final test selection and interpretation of results were made independently.

Documentation & Reporting	Claude (claude-sonnet-4-6) was the primary tool for the final report. It was used to draft, structure, and refine all major sections, including the Executive Summary, EDA write-ups, statistical analysis, business recommendations, Limitations, Future Work, Completed Work Summary, Incomplete Work Summary, Decision Log, and Reflection. ChatGPT supported the README and project documentation.
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Key Prompts Used

The following are representative examples of prompts used during the project, included to demonstrate intentionality in how AI tools were directed rather than passively accepted:

Context	Example Prompt
Data cleaning	"Help me convert a price column from string to numeric safely in pandas"
Statistical methods	"Explain when to use Mann-Whitney U test vs t-test for non-normal data"
Feature engineering	"What features can I engineer from Airbnb listing data to measure demand and revenue?"
Reporting	"Give me a professional executive summary section for a data analytics project on Airbnb London pricing"
Documentation	"Help structure a README for a data engineering project with modular Python scripts"
Statistical interpretation	"How do I interpret p-values in a business context — what does $p \approx 0$ actually mean for a stakeholder?"
Report writing (Claude)	"Write a full professional analytical report based on these Airbnb London EDA charts and statistical results, in formal tone with business interpretation for each figure"

Output Validation & Verification

No AI output was used without independent review. Validation was applied across four areas:

Validation Area	What Was Done
Code validation	All Python code was run locally; outputs were manually inspected to confirm that transformations, feature values, and chart outputs were logically correct and matched expectations from the raw data.

Analytical validation	EDA interpretations were cross-checked against the actual visualisations. AI-suggested observations were only included if they were visible and defensible in the chart itself.
Statistical validation	Test selections were confirmed against the distributional properties of the data (skewness, non-normality). Results were verified using SciPy and cross-checked for logical consistency with the exploratory findings.
Documentation validation	All report sections were reviewed for accuracy, clarity, and alignment with the assignment brief. AI-generated phrasing was rewritten where it was unclear, overly generic, or didn't reflect the actual findings of this project.

Modifications to AI-Generated Output

AI-generated content was not used as-is anywhere in this project. The main categories of modification applied were:

Type of Modification	Description
Code adaptation	AI-suggested code was rewritten to match the actual dataset structure, column names, and project-specific logic rather than used as generic boilerplate.
Feature engineering adjustments	Feature logic was refined based on observed data behaviour — for example, adjusting how occupancy was calculated after inspecting the distribution of the availability_365 field.
Report rewrites	AI-drafted explanations were rewritten to reflect genuine understanding of the findings rather than generic descriptions. Several sections went through multiple rounds of revision to ensure the business interpretation was specific to this dataset.
Simplification	Overly complex or unnecessarily verbose AI suggestions were simplified. Where AI produced lengthy academic-style prose, it was condensed to match the more direct, readable tone that was chosen for this report.
Scope alignment	Suggestions outside the prioritised scope — complex ML implementations, unnecessary libraries, multi-city frameworks — were ignored or explicitly rejected in favour of depth on the core deliverables.

Critical Assessment of AI Suggestions

A number of AI suggestions were assessed and rejected during the project. Notable examples include:

Suggestion Considered	Decision & Reason
ML-based price prediction models	Rejected — out of scope for the prioritised deliverables and would have compromised the depth of the core statistical and EDA work.
Advanced imputation techniques for missing values	Rejected in favour of median imputation and record removal — simpler approach is more reproducible and avoids introducing model-dependent bias.
Additional libraries (e.g. scikit-learn, statsmodels for regression)	Rejected — keeping dependencies minimal made the pipeline cleaner and the approach easier to validate and explain.
Multi-city comparative framework	Rejected — single-city depth was the right trade given the assignment's explicit emphasis on quality over breadth.
Complex orchestration tools (Airflow, Prefect)	Rejected — modular scripts were sufficient for the project scale and kept the pipeline readable and auditable.

Principles Governing AI Use

Throughout the project, the following principles were maintained consistently:

Principle	How It Was Applied
Full ownership of decisions	AI was consulted for options and explanations; all final calls on method selection, feature design, and interpretation were made independently.
Guidance, not answers	AI was used to understand concepts and explore approaches, not to copy final outputs. The reasoning behind each decision was understood before it was applied.
Independent validation	Every AI-generated code block, chart interpretation, and statistical claim was verified against actual data or library documentation before being included.
Understanding over automation	Where AI suggested a faster automated approach, it was only adopted if the underlying logic was understood. Black-box outputs were rejected.

Summary

AI tools — primarily ChatGPT (GPT-5) for analysis and code, Claude (claude-sonnet-4-6) for report writing, and GitHub Copilot for limited code assistance — contributed meaningfully to this project's development efficiency, code quality, analytical clarity, and documentation standard. In every case, AI outputs were reviewed, validated, and modified before use. All final decisions,

implementations, and interpretations reflect independent judgement, with AI serving as a capable and transparent collaborator rather than a substitute for critical thinking.

Conclusion

This report set out to give a structured, evidence-based look at the London Airbnb market, built on a proper data pipeline and backed up by formal statistical testing rather than just visual impressions. The analysis confirms that price here is shaped predictably by room type and neighbourhood — both statistically significant — while occupancy follows a messier, more uneven pattern that price alone doesn't explain. Superhost status comes out as a real, statistically backed signal of consistent service, and the market itself turns out to be dominated by small, individual hosts, with a smaller group of larger, likely professional, operators sitting alongside them.

Taken together, these findings give a useful starting point for the people who actually have to make decisions in this market. Hosts get pricing benchmarks broken down by segment and a clearer sense of what actually drives occupancy, rather than relying on price alone. Investors get a statistically grounded view of where prices sit geographically, to be paired with occupancy data for a fuller yield picture. The platform and anyone analysing this market more broadly can be more confident that signals like Superhost status are genuinely meaningful rather than cosmetic. The analysis has its limits, and there's plenty of room to take it further, but as it stands, it's a solid, honest starting point for thinking seriously about London's short-term rental market.